



MONASH University



Usability - S2 2025

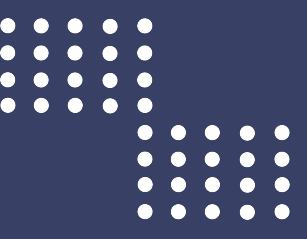
ToyTrove - A toy Sharing app

Submission 4 - Video Presentation

Google Slide Link:

https://docs.google.com/presentation/d/1JXUY4BH2Tf-zmdzNjVLikB4ZqO2RiRE2YX4u85_gvks/edit?usp=sharing





Presented By:

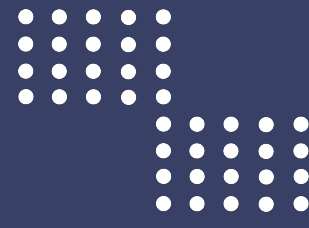
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Date : 14/11/25

Applied Class : 03

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ToyTrove



What is ToyTrove?

A mobile application that connects families to:

- Donate gently-used toys
- Swap toys with other families
- Find quality toys locally

Why ToyTrove?

- Promotes environmental sustainability
- Provides affordable toy options
- Builds stronger communities
- Ensures toys continue bringing joy

Our Goal: Reduce waste while making quality toys accessible to all families

Project Mission

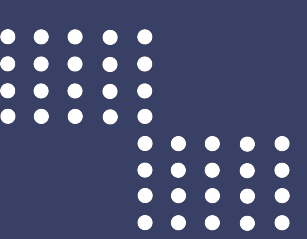


- Sustainable toy sharing and donation platform
- Connects families to donate, swap, and find quality toys locally
- Promotes environmental responsibility and community support
- Reduces waste while making toys accessible to all families

Target Audience



- Parents and caregivers with young children
- Environmentally-conscious families
- Budget-conscious households seeking affordable toys
- Community-minded individuals who finds joy in donating



Presentation Overview:

Our Evaluation Process:

- Comprehensive heuristic evaluation of high-fidelity prototype
- Identified violations across all group members
- Consolidated findings to select 4 most severe issues

Today we will present 4 critical usability violations and implemented solutions:

1. User Control & Freedom - Account Deletion
 2. User Control & Freedom - Photo Management
 3. Error Prevention - Form Validation
 4. Match Between System & Real World - Sorting Interface
- 

Severity Summary

Violation	Heuristic	Severity
Account deletion with no confirmation	H3: User Control & Freedom	4/4 (Catastrophic)
No option to delete uploaded photos	H3: User Control & Freedom	3/4 (Major)
No required field indicators	H9: Error Recognition & Recovery	3/4 (Major)
Technical sorting terminology	H2: System-World Match	3/4 (Major)

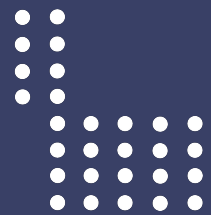
Total Impact:

- 1 Catastrophic violation
- 3 Major violations
- All violations addressed with evidence-based solutions

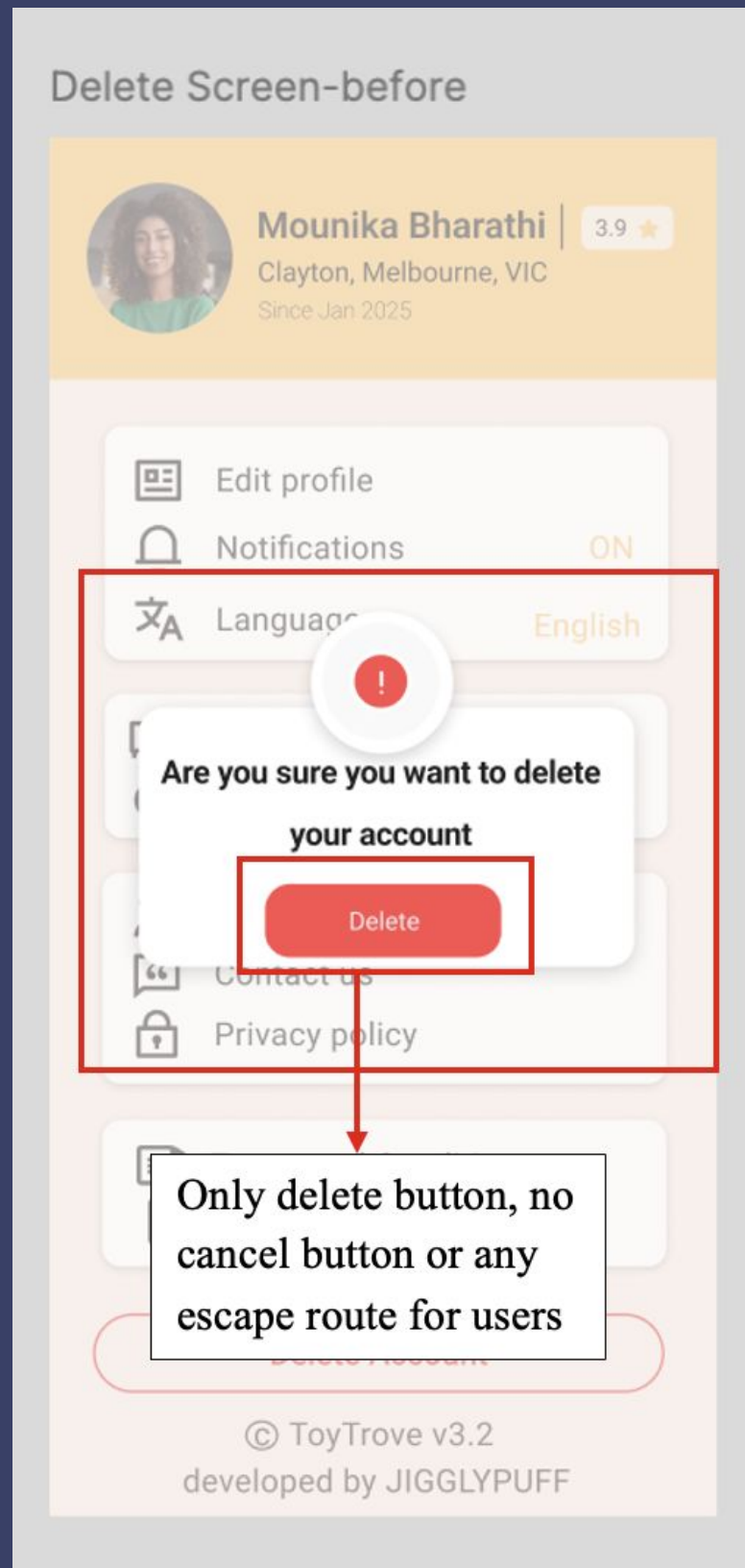


Let's Begin !

Heuristic Violations & implemented Solutions



USER CONTROL AND FREEDOM VIOLATION



PROBLEM DESCRIPTION :

- No confirmation dialog before account deletion
- One-click deletion with no warning or escape route
- Permanent loss of all account data with no recovery option

Heuristic Violated:

- Nielsen's Heuristic #3: User Control and Freedom

USER CONTROL AND FREEDOM VIOLATION

Usability Heuristic 3: User Control & Freedom

Summary: No. 3 of the top 10 UX design heuristics is to allow users freedom to be in control of the interaction, even if they make mistakes and will need a clearly marked way out of trouble.

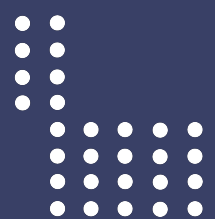
Severity: 4/4 (Catastrophic)

Frequency: 1 (Rare) - Users rarely delete accounts; typically once ever if at all.

Impact: 4 (Critical) -

- Violates Schneiderman's Golden Rule 6 (Permit Easy Reversal).
- Creates false constraint (Norman) forcing irreversible catastrophic action.
- Permanent loss of entire account, all data, listings, connections, history.
- Creates extreme anxiety. No recovery possible.

Persistence: 1 (once)- Encountered only when attempting account deletion.



Theoretical Justification

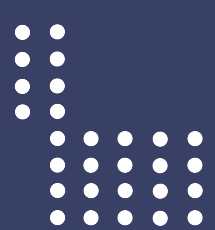
*Shneiderman's Golden Rule 6:
"Permit Easy Reversal of Actions"*

6. Permit Easy Reversal of Actions

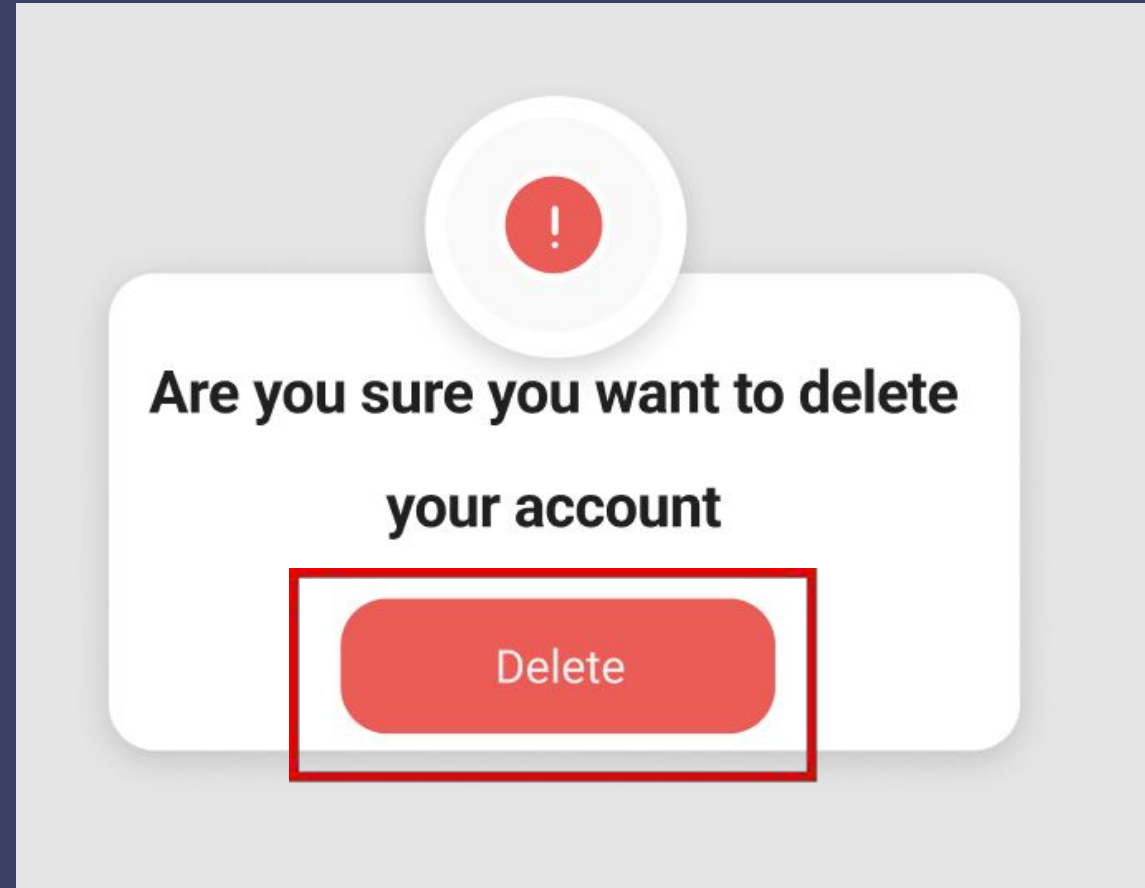
Not only for error recovery, reversible actions have psychological benefits.

Principle Explanation:

- Users should be able to reverse their actions easily
- For destructive or irreversible actions, multiple reversal opportunities must be provided
- Reduces anxiety and encourages exploration by making errors recoverable



Key Issues of the Problem



1. No Escape Route

- Once "Delete Account" is pressed, there's no way to cancel
- Users are trapped in a destructive action path
- Violates the principle of user control

2. Irreversible Consequences

- Permanent loss of all account data
- All toy listings deleted forever
- All connections and history permanently erased
- No recovery mechanism exists

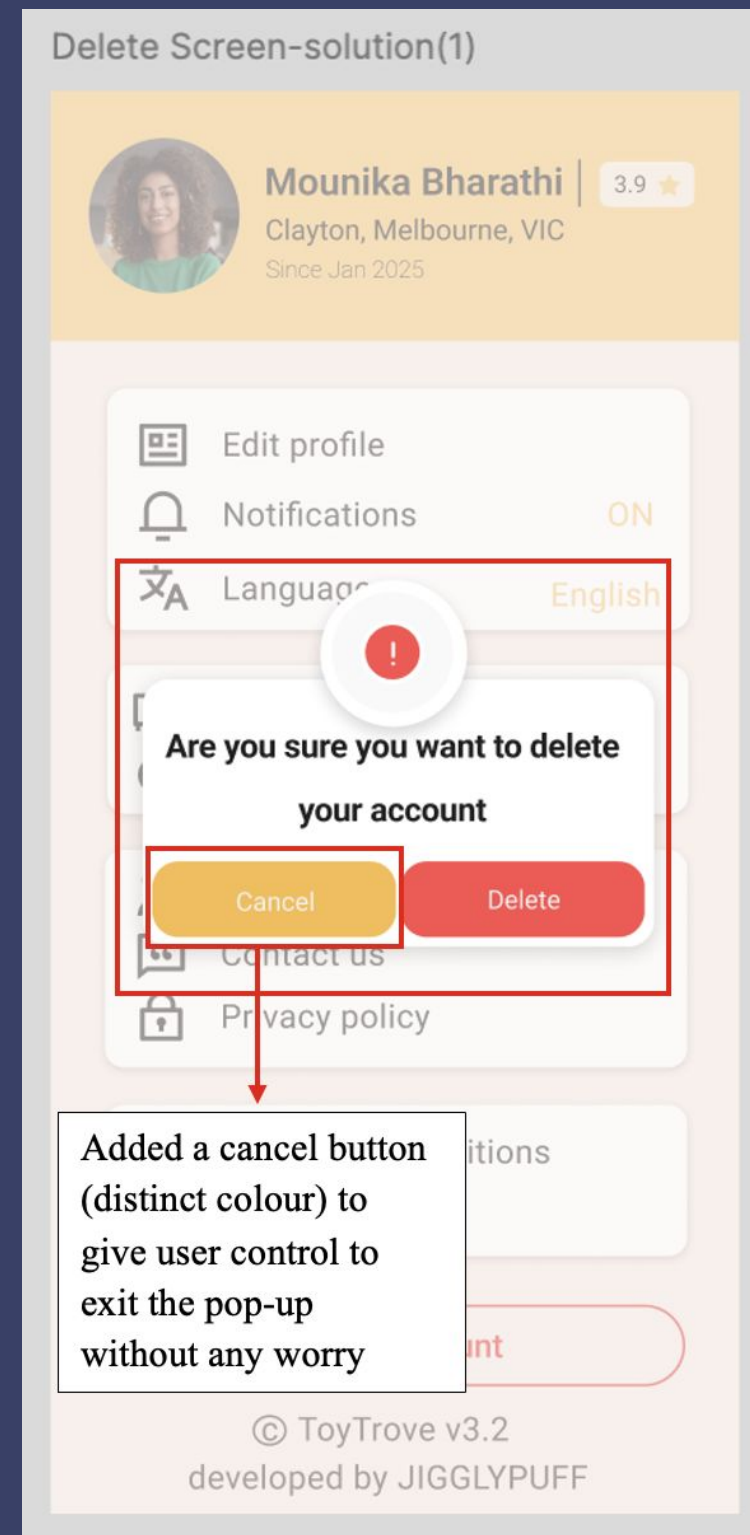
3. Lack of Warning

- No indication of the severity of the action
- Users don't understand what they're about to lose
- No explanation that deletion is permanent

4. No Safety Net

- Single point of failure - one accidental tap = disaster
- No "grace period" to undo the action
- No backup or recovery option available

Solution - First Safety Check

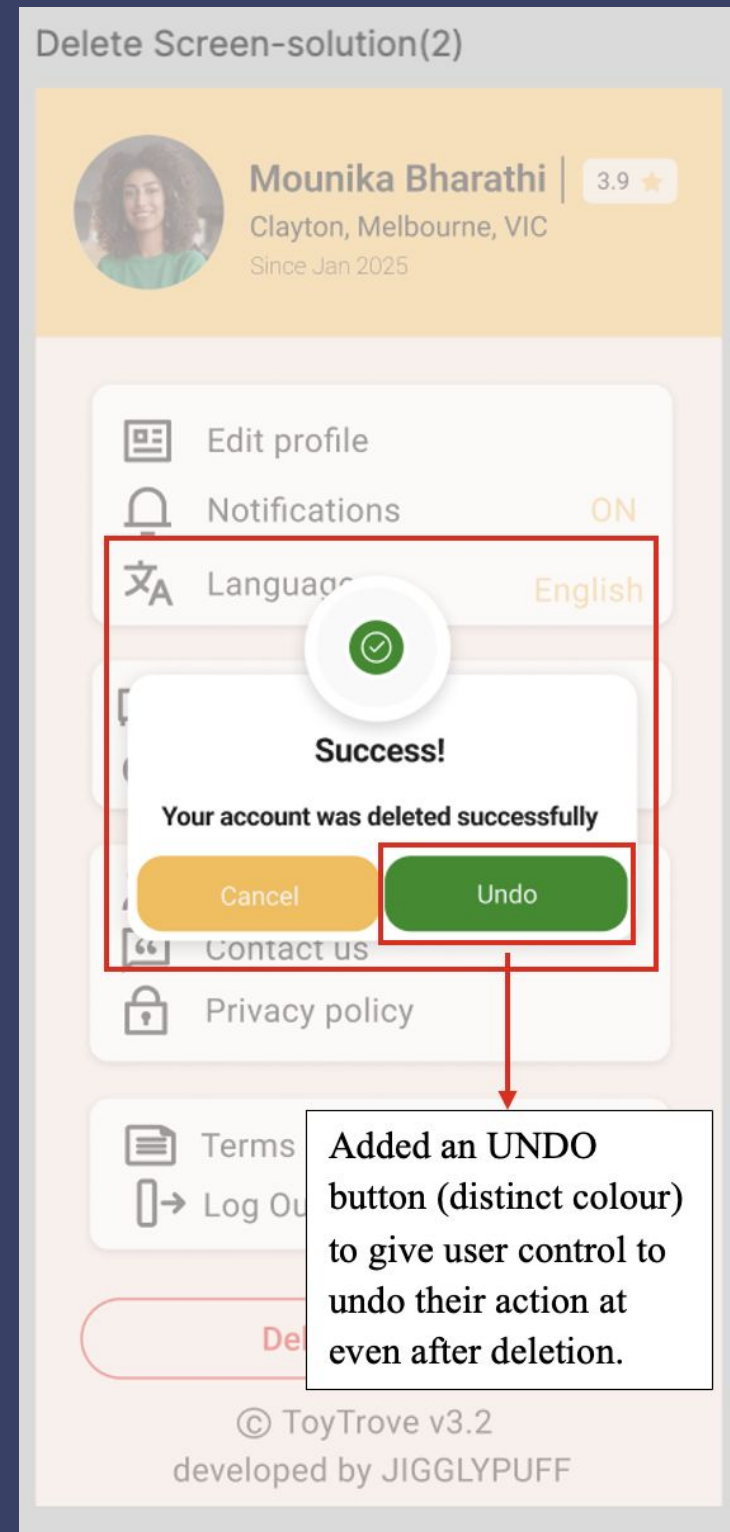


Implemented Solution:

First confirmation dialogue with clear options:

- **Warning message:** "Are you sure you want to delete your account"
- **Cancel button (left, orange):** Safe exit option
- **Delete button (right, red):** Proceeds with deletion

Solution - Second Safety Net



Additional Safety Layer:

Grace period after deletion confirmation:

- **Success feedback:** "Your account was deleted successfully"
- **Cancel button:** Alternative exit option
- **Undo button (green):** Reverses deletion immediately
- **Grace period:** Allows recovery before permanent deletion

Key Improvement: Users can reverse deletion even AFTER confirmation

Application to Solution:

Two Reversal Points Implemented:

1. First Reversal Point:

- Confirmation dialog with Cancel button
- Users can exit before deletion occurs
- Prevents accidental progression to destructive action

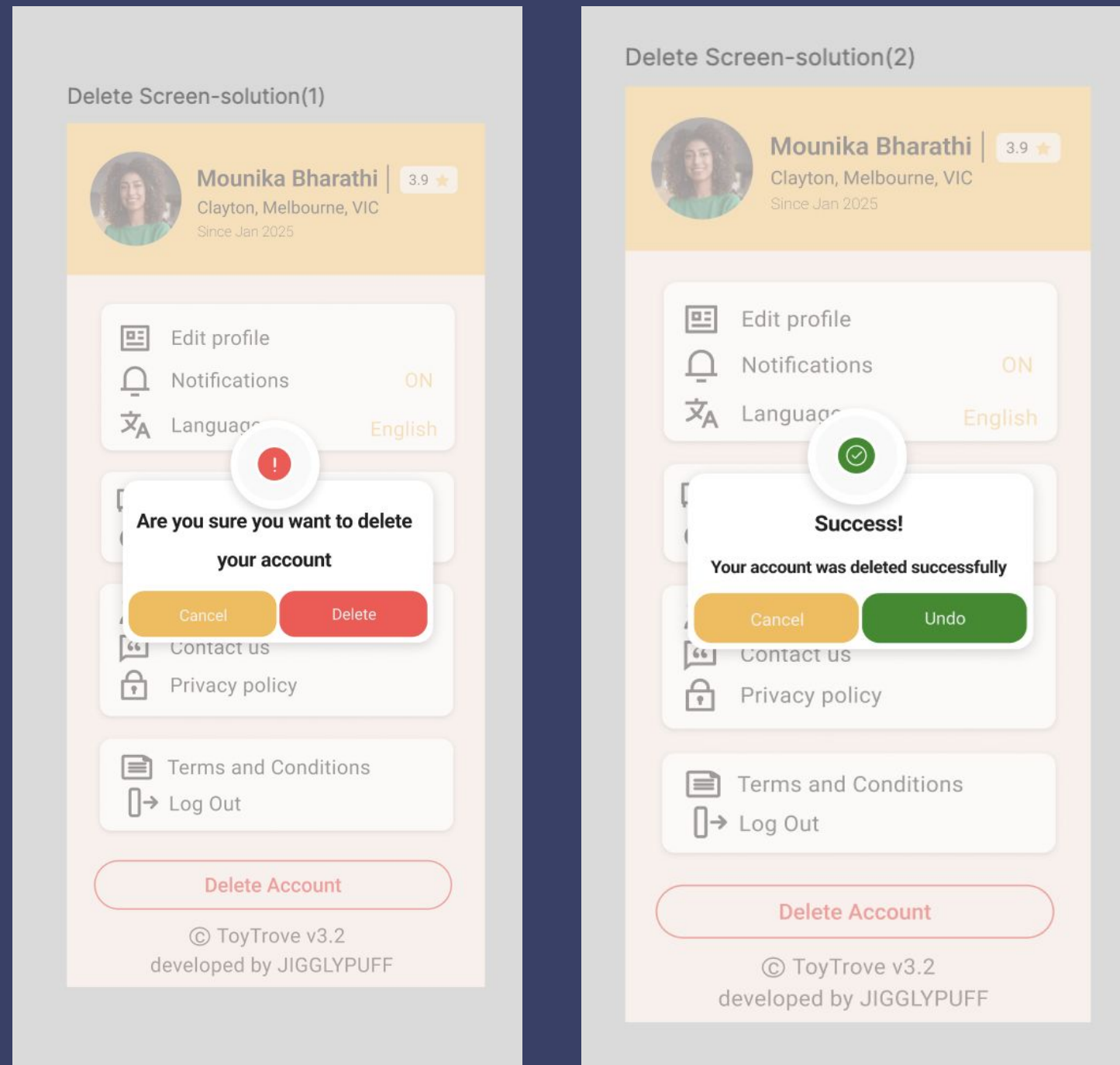
2. Second Reversal Point:

- Undo button in success message
- Grace period before permanent deletion
- Allows recovery even after confirmation

Why This Matters:

- Account deletion is catastrophic and irreversible
- Users need multiple opportunities to prevent mistakes
- Builds user confidence and trust in the system
- Reduces anxiety during high-stakes interactions

Result: Solution transforms irreversible action into a reversible process with multiple safety checkpoints



Intended Impact (ISO Usability)

ISO 9241-11 Component: Satisfaction

Definition:

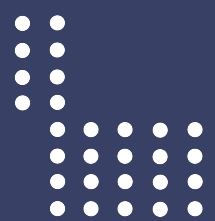
"Freedom from discomfort and positive attitudes towards the use of the product"

User Goal:

- Feel safe and in control when managing account settings
- Have the ability to reverse destructive actions if needed
- Avoid anxiety and stress during high-stakes decisions

Measure:

- User anxiety levels during account deletion process
- Sense of control and confidence ratings
- Trust in system protection mechanisms
- Psychological comfort during destructive actions



Intended Impact (ISO Usability)

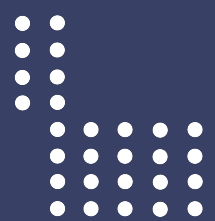
ISO 9241-11 Component: Effectiveness

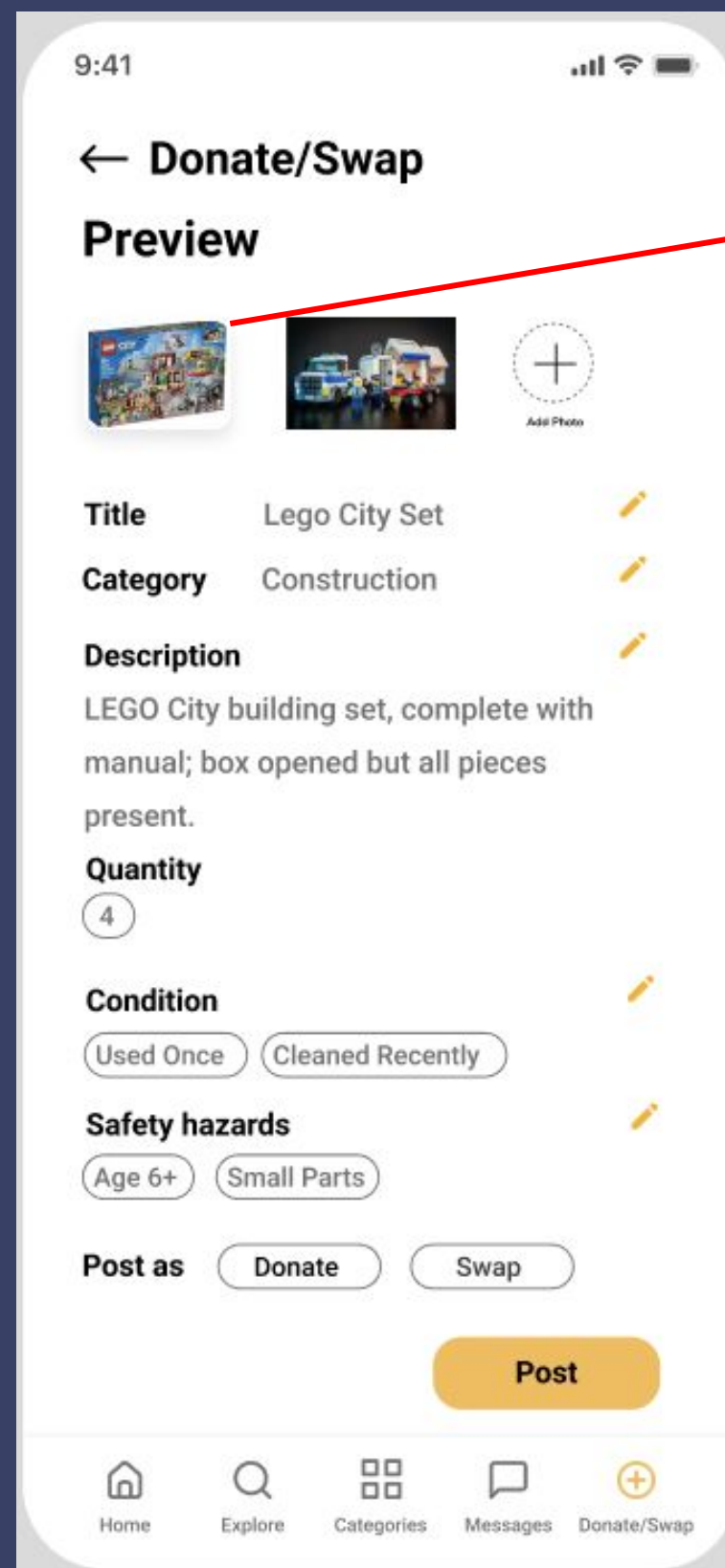
How It Works:

- **Problem:** No escape route → Users trapped → High anxiety
- **Solution:** Cancel button + Undo button → Two escape routes → Users feel safe
- **Mechanism:** Multiple reversal points eliminate feeling of being trapped

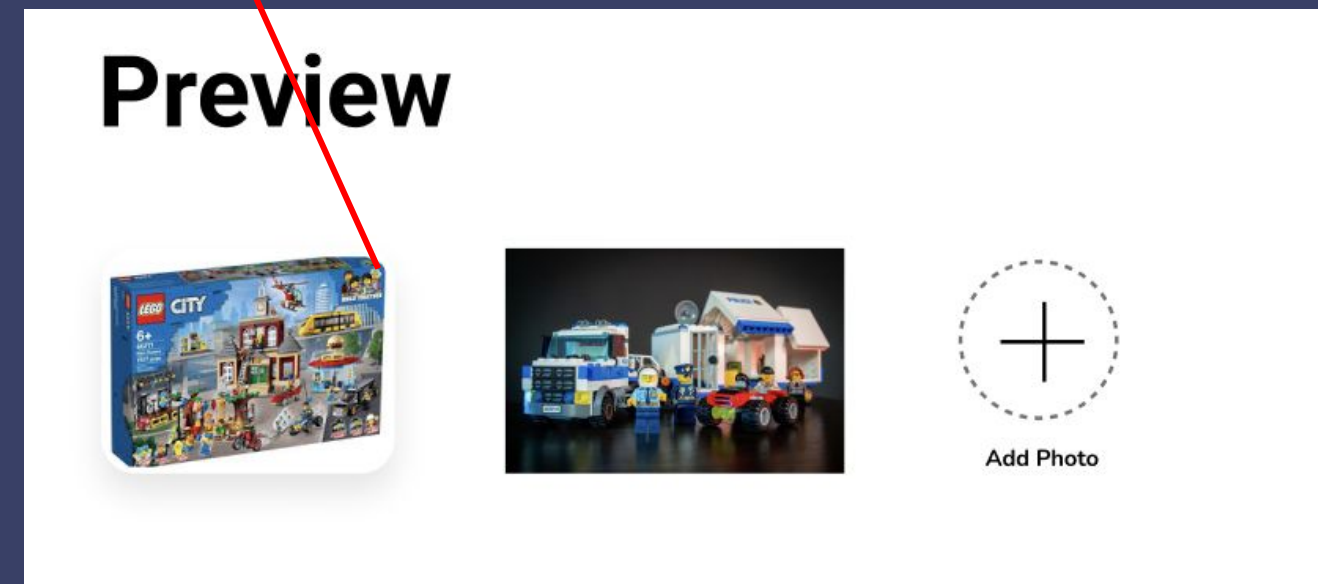
Result:

- Anxiety transformed into confidence
- Users feel protected and in control
- Trust in application increases
- Positive attitude toward account management





No Option to delete images



Heuristic 3 Violation

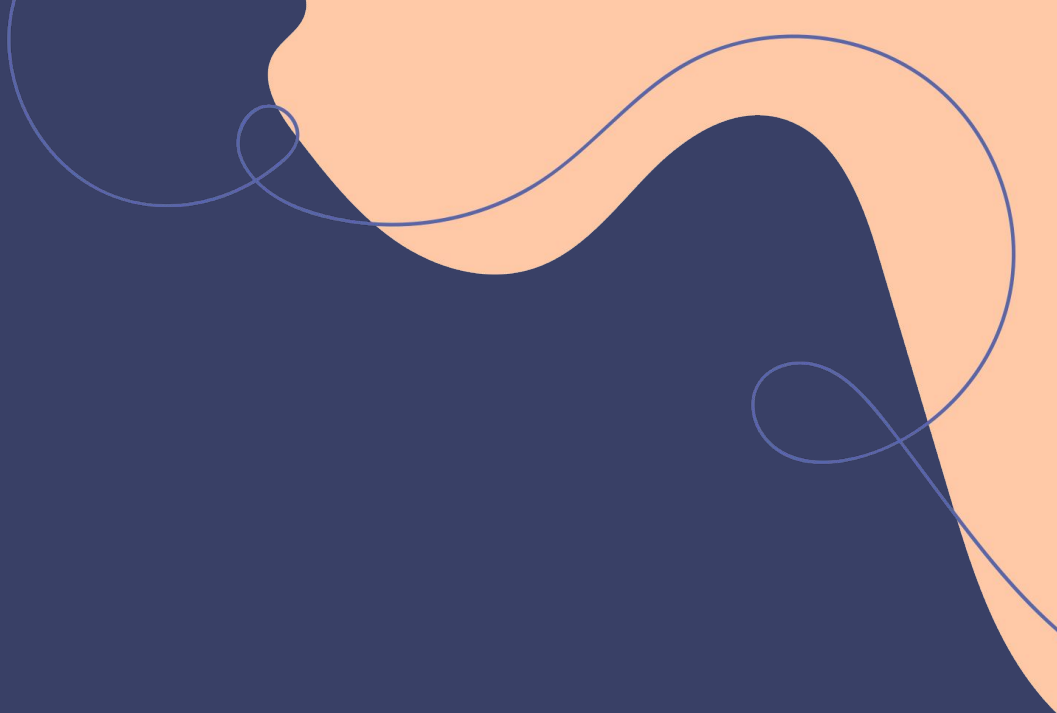
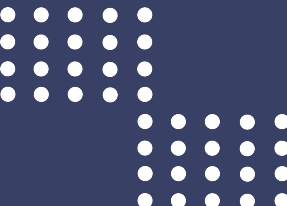
Heuristic: Lack of User Control & Freedom

Severity: 3, Major Usability Problem

Frequency: Common - users often upload wrong photos

Impact: High - forces complete restarting of listing process

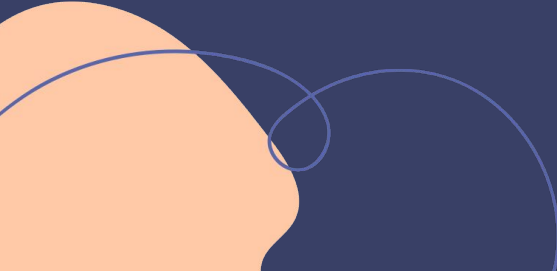
Persistence: Occurs with every incorrect photo upload



Key Issues:

1. User cannot remove incorrectly uploaded photos
2. Must abandon entire list to fix photo mistakes
3. All entered data is lost
4. Frustrates the user

Presented By: **Anish Srinath**



Preview



Solution:

1. Red X Button added to each photo thumbnail
2. One-tap deletion - Simple and fast removal
3. “Add Photo” available to immediately replace with the correct photo
4. No Confirmation Required

Preview



Additional Theories:

1. Norman's Principle
 - a. Discoverability
 - b. Affordances
 - c. Natural Mapping
2. Fitt's Law
 - a. X button large enough for accurate tapping
 - b. Predictable Position (top right corner)
 - c. Minimizes movement time



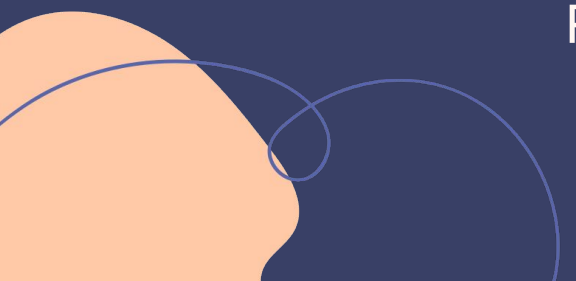
ISO Usability Impact: Effectiveness

Measure: Completion of task

Impact:

1. The user can successfully delete incorrect photos without losing other information
2. The user can upload new photo without having to go through all the steps from the beginning

Presented By: **Anish Srinath**



Title

Title of the Item

Category

Select the category

Description

Describe your toy

No required mark for form fields

Suitable for

Select the age

Quantity

Select the number of items

Review

Review looks tappable

Home Explore Categories Messages Donate/Swa

Heuristic violated: H9 — Help users recognise, diagnose & recover from errors

Severity: 3, Major Usability Problem

Frequency: Very common — occurs for every new listing with any mandatory field left blank

Impact: High — users tap **Review** repeatedly without knowing what's missing, causing trial-and-error, delays, and abandonment

Persistence: Persistent — repeats for each missing field and every attempt until the UI clearly signals requirements and blocks invalid progress

Key Issues:

- **No required indicators:** Users can't tell which fields are mandatory at the start of the form.
- **Active “Review” button:** Appears tappable even when fields are incomplete, misleading users.

The screenshot shows a form with the following fields:

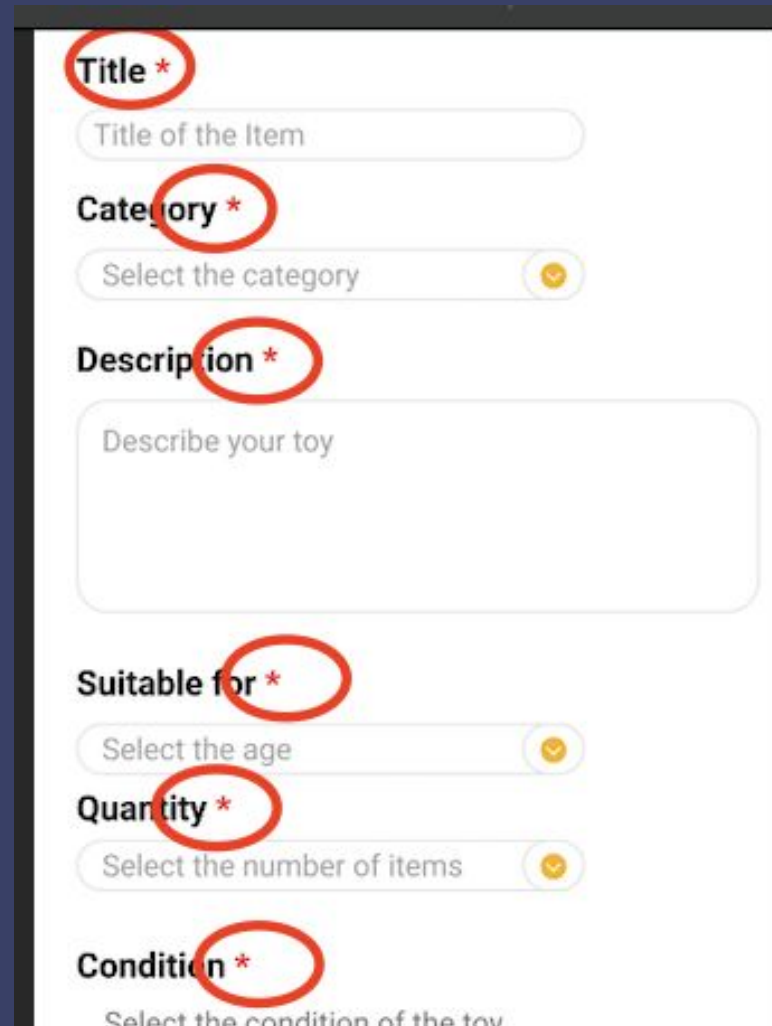
- Title**: Input field with placeholder text "Title of the Item".
- Category**: Dropdown menu with placeholder text "Select the category".
- Description**: Text area with placeholder text "Describe your toy". A red arrow points to this field with the text "No required mark for form fields".
- Suitable for**: Dropdown menu with placeholder text "Select the age".
- Quantity**: Dropdown menu with placeholder text "Select the number of items".

The screenshot shows a mobile app interface with a list of items and a prominent yellow "Review" button. A red arrow points to the button with the text "Review looks tappable". The list items include:

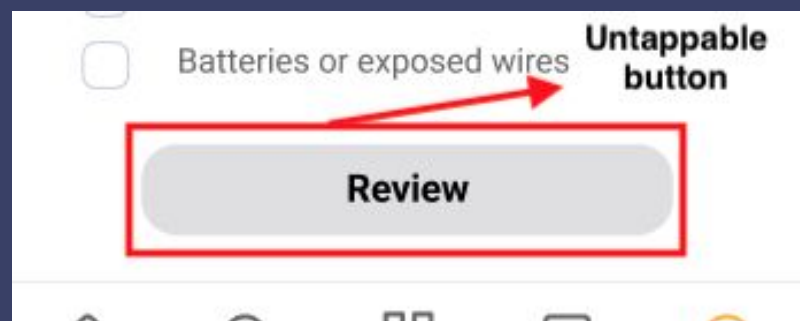
- ☐ Detachable small parts
- ☐ Batteries or exposed wires

The bottom navigation bar includes icons for Home, Explore, Categories, Messages, and Donate/Swa.

Presented By: **Rachit Rajiv**



A screenshot of a mobile application form for adding a toy. The form contains several input fields, each with a red circle around an asterisk (*) indicating it is a required field. The fields are: 'Title *' (text input), 'Category *' (dropdown menu), 'Description *' (text area), 'Suitable for *' (dropdown menu), 'Quantity *' (dropdown menu), and 'Condition *' (dropdown menu). The text 'Select the condition of the toy' is partially visible at the bottom.



Solution:

1. * **Added “Required”** indicators beside all mandatory fields to clearly signal what must be filled.
2. **Disabled the “Review” button** until all required fields are completed — shown in **grey** to indicate it's not tappable.
3. **Button activates automatically** once all mandatory inputs are valid, giving immediate feedback.

Title *

Title of the Item

Category *

Select the category

Description *

Describe your toy

Suitable for *

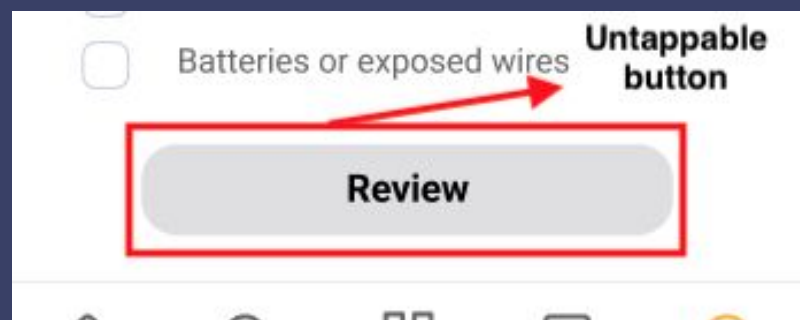
Select the age

Quantity *

Select the number of items

Condition *

Select the condition of the toy



Why the solution works

Heuristic : H9

Solution follows Norman's Constraints :

Constraints **block invalid actions** and guide users to the allowed path, reducing error and memory load.

Application:

- **Disabled “Review” button** acts as a **constraint**, preventing users from progressing with incomplete data.
- **“Required” indicators* serve as **signifiers**, clearly showing what inputs are necessary before continuing

ISO Usability Impact - Effectiveness

ISO Usability Component: Effectiveness — users can complete the “Create Listing” task accurately and fully.

Intended Impact:

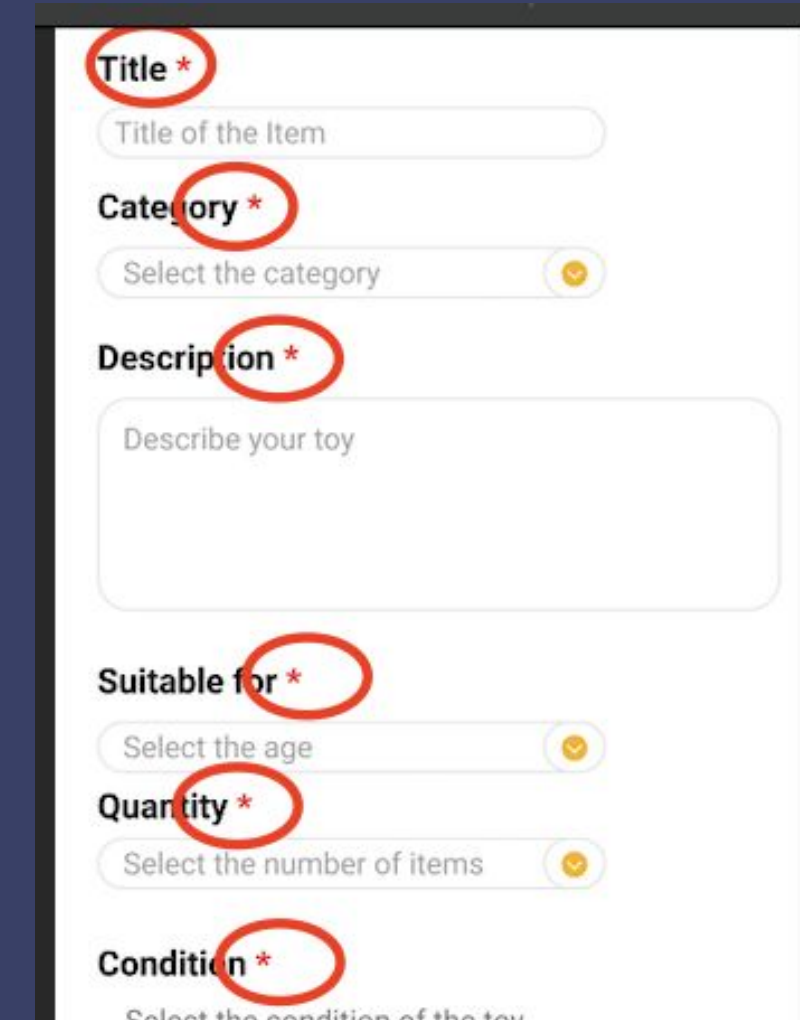
Required indicators **guide users** by showing what must be filled.

Disabled **Review** button **constraints invalid actions**, preventing early submission.

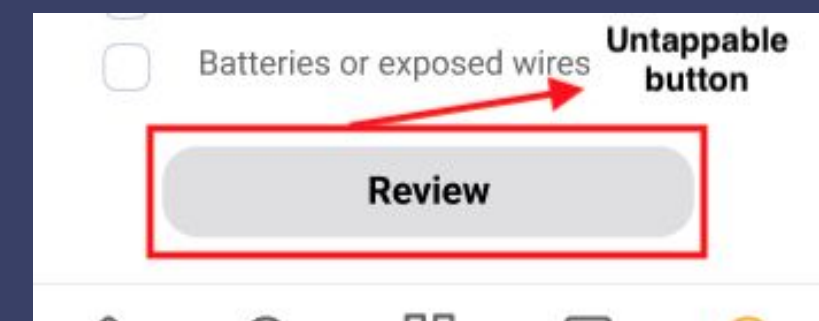
Relevant Measure:

- **First-pass completion rate** (no missing-field errors before Review) — expected increase of ~25%.
- **Number of error corrections needed** per attempt — expected reduction of ~50%.

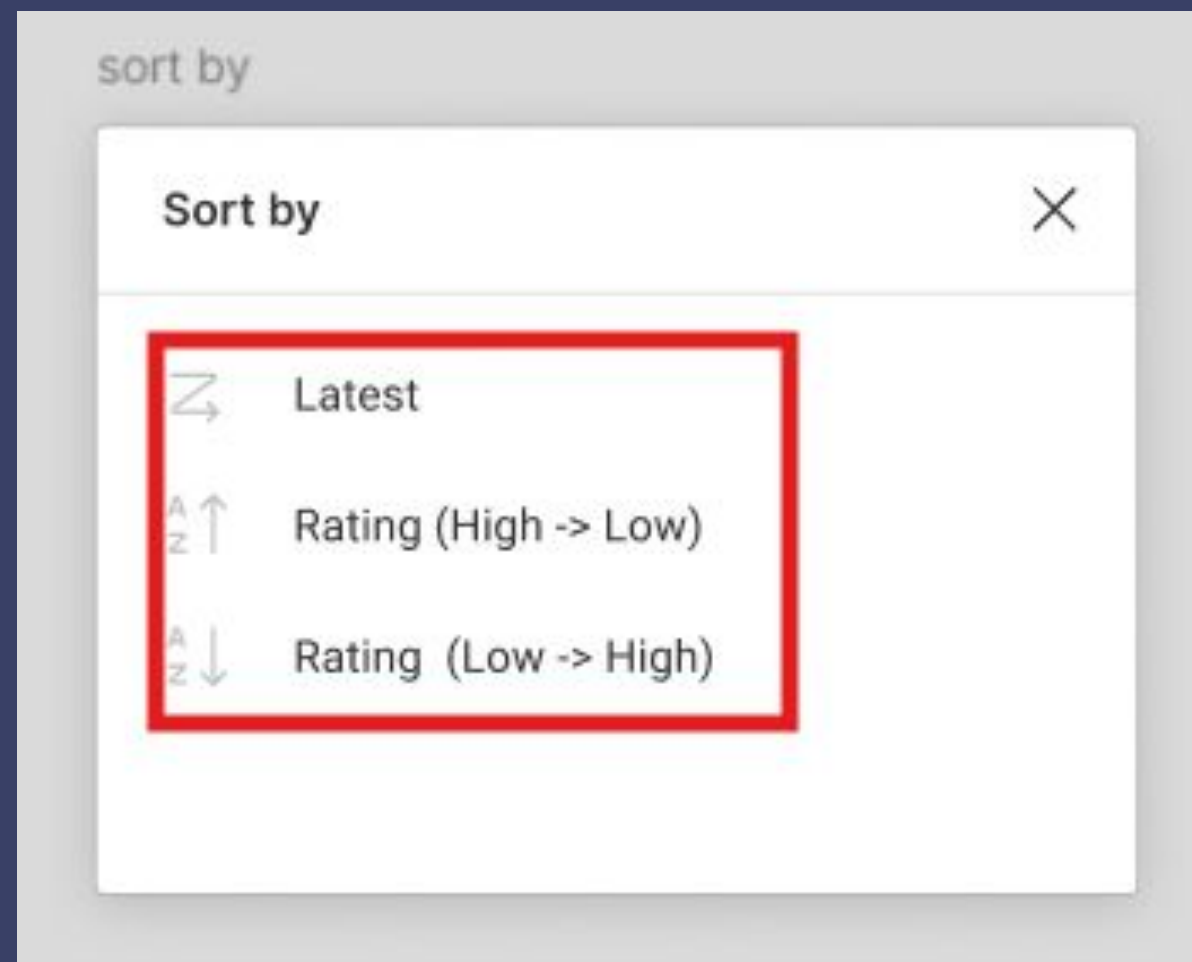
Presented By: **Rachit Rajiv**



A screenshot of a mobile application form titled 'Create Listing'. The form contains several input fields, each with a red circle around its label and an asterisk to indicate it is required. The fields are: 'Title *' (with placeholder 'Title of the Item'), 'Category *' (with placeholder 'Select the category'), 'Description *' (with placeholder 'Describe your toy'), 'Suitable for *' (with placeholder 'Select the age'), 'Quantity *' (with placeholder 'Select the number of items'), and 'Condition *' (with placeholder 'Select the condition of the toy').

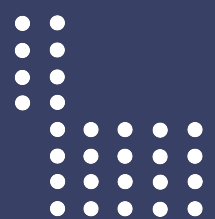


Heuristic violated: H2: Match Between System and Real World



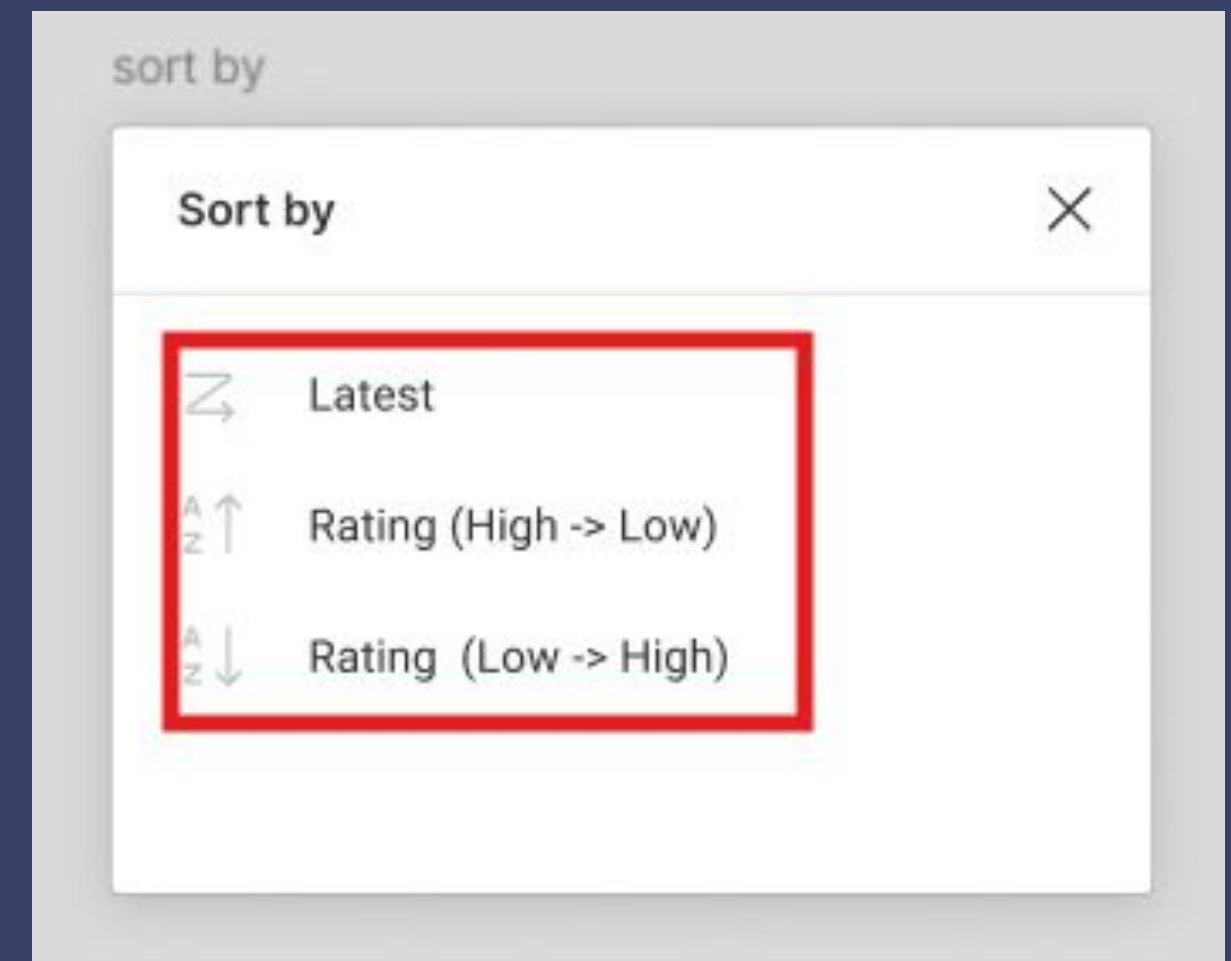
PROBLEM DESCRIPTION :

- **Severity: 3**
- **Frequency: High** - This will affect everyone who tries to sort their search results, so it's quite common.
- **Impact: Moderate** - confuses sorting criteria, may lead users to avoid the sorting feature entirely
- **Persistence: Repeated** - occurs every time users want to organise their search.
- The system should speak the user's language using familiar words and concepts
- Instead of technical terms, interfaces should follow real-world conventions

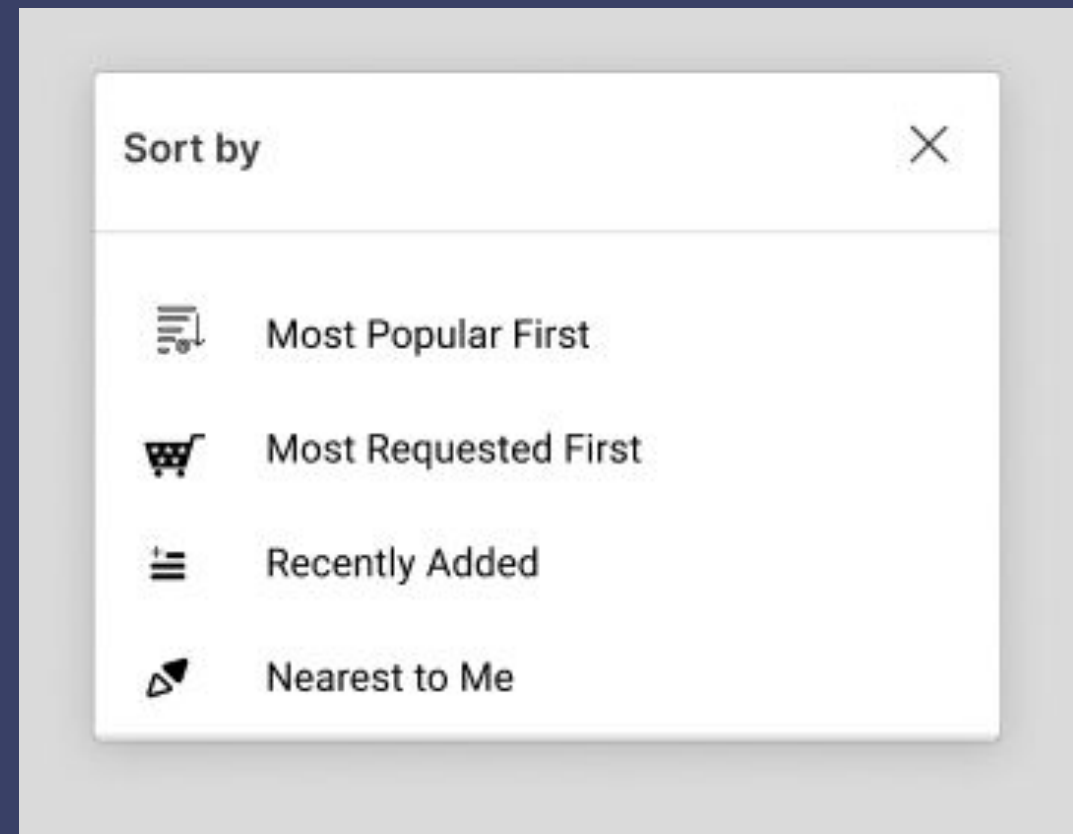


Key Issues:

- Sort menu used technical terminology that didn't match how users think
- Terms like "Rating (High > Low)" were unclear and abstract
- The ">" symbol felt like coding language, not everyday speech
- Users had to guess what "rating" actually meant in this context



Solution Implemented:

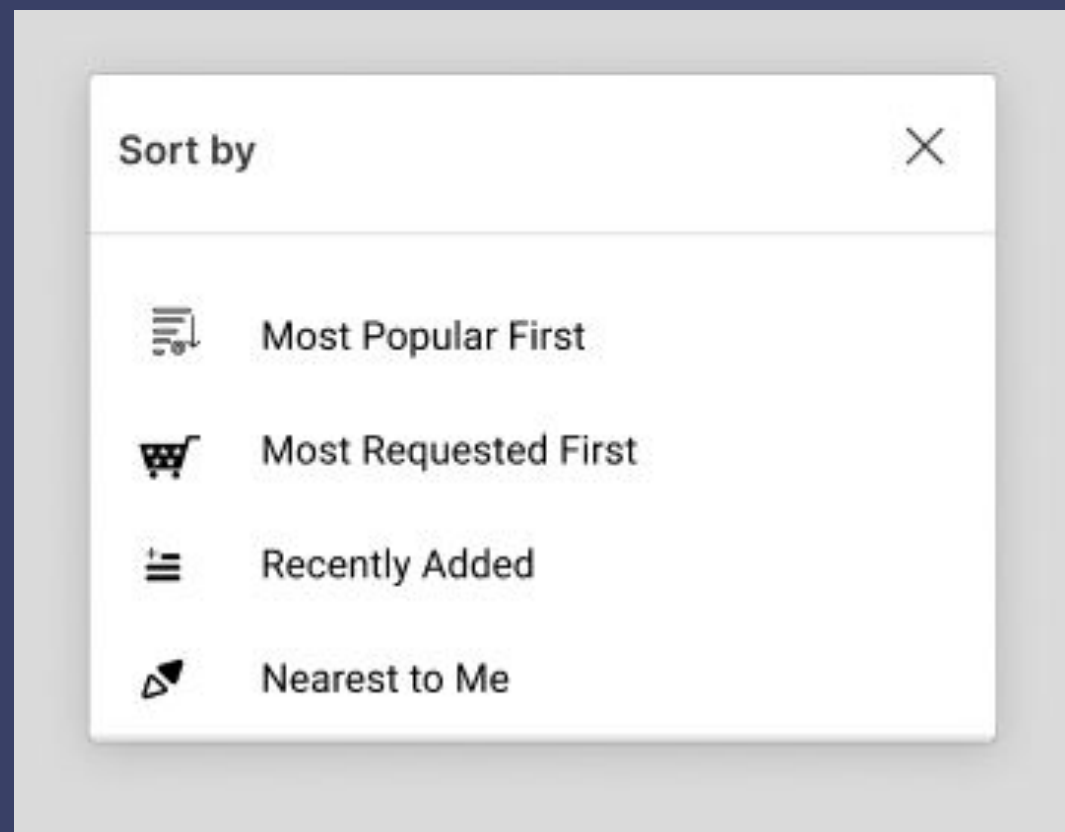


1. **"Most Popular First"** - clear, user-friendly language
2. **"Most Requested First"** - describes actual user behavior
3. **"Recently Added"** - specific and understandable
4. **"Nearest to Me"** - relates to user's physical context
5. Added **icons** alongside each option (star, hand, clock, location pin)
6. Removed all technical symbols and jargon

Why the solution works

Norman's Principle of Natural Mapping

- Controls should match users' existing mental models
- New labels reflect how toy sharers actually think:
 - **"Most Popular"** : what other families choose
 - **"Most Requested"** : What toys are most in demand
 - **"Recently Added"** : temporal freshness
 - **"Nearest to Me"** : geographic proximity
- Replaces abstract "rating" with concrete, familiar terms
- Makes the relationship between option and result immediately obvious

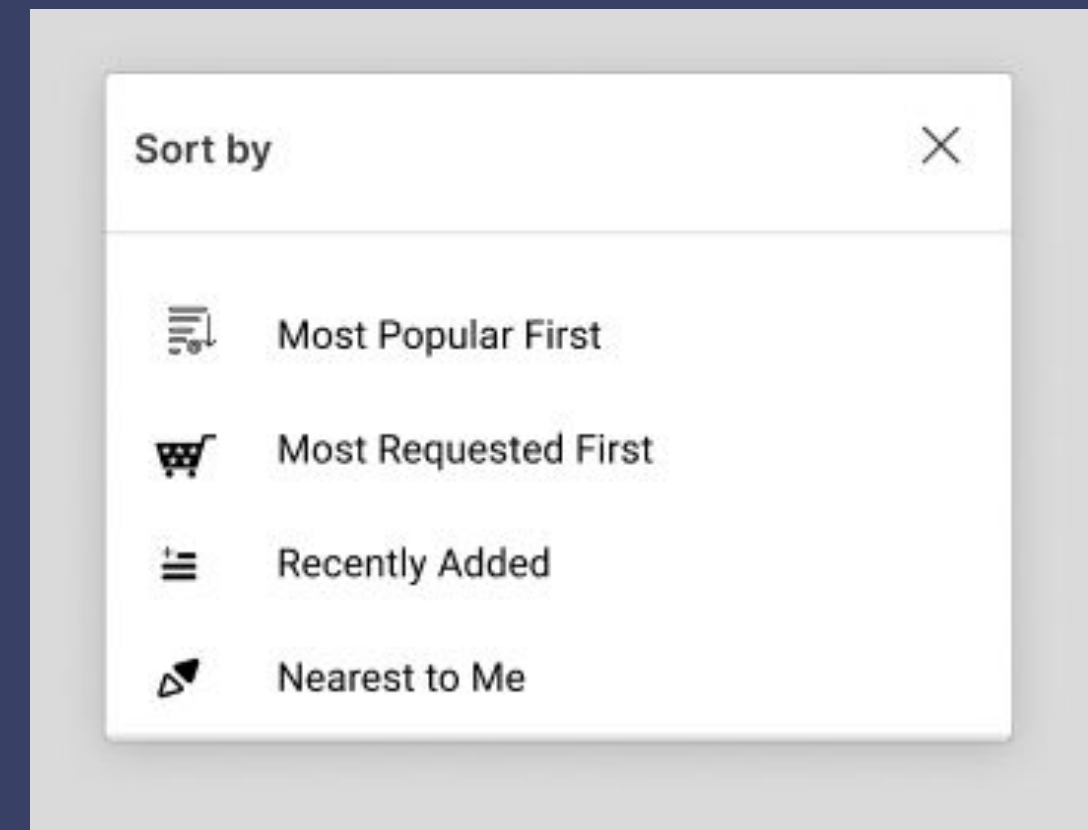


ISO 9241-11 Component: Effectiveness

Measure: Task completion rate

Expected Impact:

- Users successfully organize results on first attempt
- Eliminates guessing and trial-and-error behavior
- Familiar terminology removes cognitive barriers
- Users confidently select the exact sorting method needed
- Icons provide secondary visual cues for comprehension



References

1. Nielsen, J. (1994). *10 Usability Heuristics for User Interface Design*. Nielsen Norman Group. Retrieved from <https://www.nngroup.com/articles/ten-usability-heuristics/>
2. Shneiderman, B., Plaisant, C., Cohen, M., Jacobs, S., Elmqvist, N., & Diakopoulos, N. (2016). *Designing the User Interface: Strategies for Effective Human-Computer Interaction* (6th ed.). Pearson Education.
3. ISO 9241-11:2018. *Ergonomics of human-system interaction — Part 11: Usability: Definitions and concepts*. International Organization for Standardization.



Thank you so much